



CROWD FUNDING PLATFORM BY USING BLOCKCHAIN

A PROJECT REPORT

Submitted by

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in partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING

**KONGUNADU COLLEGE OF ENGINEERING AND TECHNOLOGY
(AUTONOMOUS)**

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**KONGUNADU COLLEGE OF ENGINEERING AND TECHNOLOGY,
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PEO III: Graduates shall have professional ethics, team spirit, lifelong learning, good oral and written communication skills, and adopt the corporate culture, core values and leadership skills.

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2. **Problem analysis:** Identity, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using the first principles of mathematics, natural sciences, and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety, and the ural, societal, and environmental considerations.
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8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
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10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-Long Learning:** Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

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BONAFIDE CERTIFICATE

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INTERNAL EXAMINER

EXTERNAL EXAMINER

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ABSTRACT

Crowdfunding is a method of online fundraising process that was initially developed for public members. It is to make contributions to support the projects of creative individuals. The integration of smart contracts ensures a secure, transparent, and safe fundraising environment. This innovation aims to enhance campaign development and financial contributions through interactive forms, providing visibility into campaign progress for both creators and donors. The current global crowdfunding market faces challenges with campaigns lacking transparency, leading to fraudulent activities and unreliable crowd investment efforts. These smart contracts facilitate a secure, transparent, and trustless environment for the platform's operations. Blockchain inherent security features, such as immutability and decentralized mechanisms that are utilized to ensure the integrity and transparency of all transactions and interactions within the crowdfunding platform. This security algorithm, leveraging the power of Blockchain and Ethereum smart contracts, aims to eliminate fraud, provide transparency in campaign management, and build trust among participants by offering a tamper-proof and automated system for crowdfunding operations.

Keywords: Crowdfunding, Blockchain, Ethereum smart contracts, Campaigns, Secure.

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LIST OF ABBREVIATIONS

CSS	Cascading Style Sheets
DApps	D ecentralized A pplications
DAO	D ecentralized A utonomous O rganizations
DeFi	D ecentralized F inance
ER	E ntity R elationship
HTML	H yper T ext M arkup L anguage
ICO	I nitial C oin O ffering
IPFS	I nter P lanetary F ile S ystem
UI	U ser I nterface
SCP	S ocially oriented C rowdfunding P latforms

CHAPTER 1

INTRODUCTION

Crowdfunding platforms are online centers where individuals go to get cash for their initiatives or ideas. Their platforms facilitate the connection between creators and small-contributing backers to support their activities. Donation-based, reward-based, equity-based, and debt-based are some of the varieties. Artists present their ideas, outline their financial objectives, and describe how money will be used. An innovative notion in finance, the model allows backers to get rewards or shares based on the success of the venture. This has enabled many entrepreneurs and innovators to realize their ideas by utilizing the mass population's combined might.

Crowdfunding has entered a new era of efficiency, security, and transparency with the introduction of blockchain technology. A blockchain-based crowdfunding platform takes advantage of the decentralized structure of blockchain technology to solve a number of issues that arise with conventional crowdfunding models.

Blockchain-based crowdfunding systems provide unmatched confidence and security to project creators and backers through the use of smart contracts, immutable ledgers, and decentralized consensus mechanisms. Moreover, blockchain-based crowdfunding platforms maximize efficiency and lower administrative expenses by doing away with middlemen and automating procedures with smart contracts. This allows more money to go straight to project creators.

This democratization of fundraising democratizes access to capital, opening up opportunities for innovators and creators who may have been overlooked or excluded by traditional financial institutions. In summary, a crowdfunding platform built on blockchain technology represents a transformative evolution in the way projects are funded and supported.

By combining the power of crowdfunding with the security and transparency of blockchain, these platforms have the potential to unlock new avenues of innovation, foster global collaboration, and drive positive change across industries. In recent years, crowdfunding has become a popular method for raising capital. However, traditional platforms face challenges like transparency and security. Blockchain technology offers a decentralized solution, revolutionizing crowdfunding.

Blockchain, known for cryptocurrencies like Bitcoin and Ethereum, creates a distributed ledger system. This eliminates the need for intermediaries, making transactions more efficient and secure.

Blockchain crowdfunding platforms offer transparency through immutable ledgers and smart contracts. They also provide global accessibility, reaching a wider audience without geographical restrictions. Tokenization allows assets to be represented as digital tokens, enabling fractional ownership and easier transferability.

By reducing fees and intermediaries, blockchain crowdfunding democratizes access to capital, particularly for small-scale projects. However, challenges remain, including regulatory compliance, scalability, and user adoption.

Despite these challenges, blockchain-based crowdfunding platforms have the potential to reshape fundraising, fostering innovation and inclusivity in the financial ecosystem. Moreover, blockchain-based crowdfunding platforms eliminate the need for middlemen, reducing administrative expenses and maximizing efficiency. With automated processes and decentralized consensus mechanisms, blockchain platforms streamline the fundraising process, allowing more capital to flow directly to project creators. This not only benefits project creators by providing them with greater access to funding but also empowers backers to support projects with confidence, knowing that their contributions are being utilized as intended.

1.1 OBJECTIVE

- Implementing blockchain technology aims to bring unprecedented transparency to the crowdfunding process.
- Security is a significant concern in crowdfunding, with instances of data breaches and fund misappropriation.
- Traditional crowdfunding platforms often have limitations regarding who can participate based on geographical location or financial infrastructure.

1.2 OVERVIEW

This project aims to develop a crowdfunding platform using blockchain technology provides a decentralized system for individuals and businesses to raise funds for projects. It leverages blockchain's transparency, security, and global accessibility to streamline the crowdfunding process. Users can create campaigns, receive contributions, and manage funds through smart contracts, ensuring transparency and reducing the need for intermediaries. Tokenization of assets allows for fractional ownership and easier transferability. Overall, it aims to democratize access to capital, support innovation, and empower both project creators and backers.

1.3 BLOCKCHAIN

Blockchain technology revolutionizes data management by offering a decentralized, transparent, and secure platform for recording transactions. At its core, blockchain comprises a chain of blocks, each containing a timestamp and a link to the previous block, ensuring the integrity and immutability of the data. This decentralized ledger system operates across a network of computers (nodes), eliminating the need for a central authority. Transactions are validated through consensus mechanisms like proof of work or proof of stake, ensuring trust and security. Blockchain's applications extend beyond cryptocurrencies to various industries, including finance, supply chain management, healthcare, and voting systems, where it enhances transparency, efficiency, and security. Smart contracts

further automate processes by executing predefined actions when specified conditions are met. Despite its potential, blockchain faces challenges such as scalability and regulatory concerns. Nevertheless, ongoing research and development aim to address these issues, paving the way for blockchain's continued evolution and adoption across diverse sectors.

As blockchain technology continues to mature, its impact on society becomes increasingly profound. The immutable and transparent nature of blockchain ensures that historical data remains intact and verifiable, fostering trust in transactions and processes. Moreover, its decentralized architecture empowers individuals by granting them greater control over their data and assets. This democratization of information has the potential to reshape power dynamics in various industries, promoting inclusivity and reducing reliance on centralized authorities. While challenges persist, such as scalability and interoperability, ongoing innovation and collaboration within the blockchain ecosystem promise to overcome these obstacles. Ultimately, blockchain represents a paradigm shift in how we manage and exchange value, paving the way for a more transparent, efficient, and equitable future.

As blockchain technology matures, its potential for disrupting traditional systems becomes increasingly evident. From streamlining supply chain operations to enhancing financial transactions, blockchain offers versatile solutions to complex problems. Moreover, its decentralized nature reduces the risk of single points of failure and censorship, bolstering resilience and trust in critical infrastructure. With continued research and development, coupled with growing adoption across industries, blockchain is poised to revolutionize the way we interact, transact, and collaborate in the digital age.

CHAPTER 2

LITERATURE SURVEY

[1] TITLE: CROWD - FUNDING USING BLOCKCHAIN (2023).

AUTHOR: Prof. Sonali Dongare, Vishwajeet Bamane, Abhishek More, Prasad Chopade, Saurabh Aherwadkar, Nitin Dhawas.

This review paper explores the transformative potential of blockchain-based crowdfunding, addressing the vulnerabilities of traditional platforms, including data breaches and high fees. Analyzing three research papers focused on Ethereum-based donation transactions, the study emphasizes the strengths and limitations of these platforms. While blockchain introduces transparency, trust, and security, challenges such as technical complexity, volatility, and scalability persist. Smart contracts and digital identity management emerge as key methodologies to enhance privacy and decentralization. The paper highlights the global accessibility, reduced intermediaries, and immutable transactions afforded by blockchain. Despite challenges, this synthesis underscores blockchain's role in revolutionizing crowdfunding, offering insights for future research and practical applications. Against this backdrop, the paper aims to provide a comprehensive review of existing literature, offering insights into the strengths, limitations, and user perceptions surrounding blockchain-based crowdfunding. The research also proposes a blockchain-centric crowdfunding network designed to confer privacy, security, and decentralization, emphasizing the transformative potential of Ethereum's smart contracts in reshaping the crowdfunding paradigm.

TECHNIQUES: Smart Contracts, Transparency, Cryptocurrency, Tokenization ,
Ethereum.

MERITS: The distributed and immutable nature of blockchain ensures that recorded transactions cannot be easily altered.

[2] TITLE: CROWDFUNDING PLATFORM BY USING SMART CONTRACT (2023).

AUTHOR: Raunak Sulekh , Manas Katiyar , Devang Trivedi.

This study examines the use of blockchain technology to construct a crowdfunding platform. The opening section of the article looks at the current state of crowdfunding and the shortcomings of conventional platforms, including transparency, security and centralization concerns. In this paper, the architecture and functionality of a crowdfunding platform built on a blockchain network are examined as a case study. The platform enables safe and transparent fundraising, money tracking, and incentive delivery. The findings of this study imply that a blockchain-based crowdfunding platform can overcome many of the drawbacks of conventional crowdfunding platforms and offer a more safe and transparent method of fundraising after thorough investigation of the literature and case studies. The goal of using Ethereum smart contracts on the crowdfunding platform is to assure automatic contract execution to stop fraud and promote confidence between fundraisers, donors and the platform. Smart Contracts will be developed to handle transaction and administrative tasks within the application on the Blockchain network, showcasing the practical workings of web3, blockchain, Solidity and smart contracts.

TECHNIQUES: Smart Contracts, Transparency, Cryptocurrency, Tokenization ,
Ethereum.

MERITS: Uses Blockchain-based crowdfunding platforms leverage smart contracts for automated operations, enhancing transparency and security.

[3] TITLE: DECENTRALIZED CROWDFUNDING PLATFORM USING BLOCKCHAIN (2023).

AUTHOR: Arohi Rathore, Syed Wajahat Abbas Rizvi.

This thesis delves into the potential of Blockchain technology and the power of Smart Contracts in crafting Decentralized Applications (DApps), Decentralized Autonomous Organizations (DAO), and advancing Cryptocurrency development for asset and wealth exchange. Smart Contracts will be developed to handle transaction and administrative tasks within the application on the Blockchain network, showcasing the practical workings of web3, blockchain, Solidity and smart contracts. The foundational elements of this decentralized crowdfunding platform involve linking a React application with the blockchain and an Ethereum wallet through MetaMask. MetaMask, as a Chrome plugin, serves as a secure wallet, facilitating communication with our decentralized application. It allows users, through a compatible web browser or the app's embedded browser, to securely connect to Decentralized Applications (DApps). These DApps, built on the Ethereum blockchain, encompass a wide array of services and functionalities, ranging from Decentralized Finance (DeFi) platforms to digital asset exchanges and Decentralized Autonomous Organizations (DAOs). By interfacing with these DApps, users can participate in various activities such as lending and borrowing assets, trading digital tokens, accessing decentralized marketplaces, and engaging in governance processes.

TECHNIQUES: Tokenization, Smart Contracts, Transparency, Cryptocurrency, Ethereum, Decentralized Applications.

MERITS: The content dives into technical details like React integration, MetaMask functionality, and Solidity coding, demonstrating a thorough understanding of the subject matter.

[4] TITLE: CROWD-FUNDING USING BLOCKCHAIN TECHNOLOGY (2022).

AUTHOR: Prof D.L.Faak, Shanawaz ,Jadhav Pranav, Kakte , Shukla Utkarsh.

Crowdfunding, initially conceived as a means for the public to support creative projects by contributing small amounts of money, has evolved into a method for individuals to invest in startups through intermediaries like broker-dealers. However, existing crowdfunding platforms lack a Donor Guarantee Policy and provide limited control over contributed funds. This article proposes the utilization of blockchain technology to address these shortcomings and create a safe, secure and transparent crowdfunding platform. By leveraging blockchain technology, the platform can offer enhanced security and transparency. Blockchain's immutable ledger ensures that all transactions are recorded transparently and securely, providing donors and campaign creators with greater confidence in the integrity of the platform. Furthermore, blockchain technology enables real-time tracking of funds, ensuring that donors can monitor the flow of their contributions. All transactions are recorded and stored in blocks on the blockchain, providing a permanent and tamper-proof record of financial activities. In conclusion, by harnessing the capabilities of blockchain technology, this paper aims to establish a crowdfunding platform that offers enhanced security, transparency, and user-friendliness.

TECHNIQUES: Ledger ,Tokenization, Smart Contracts, Transparency, Cryptocurrency, Ethereum.

MERITS: It offers enhanced security and transparency, breaking free from centralized control. They provide precise inputs for investors, such as expected returns and exit periods, fostering informed investment decision.

[5] TITLE: ROLE OF BLOCKCHAIN TECHNOLOGY CROWD FUNDING (2021).

AUTHOR: Loan T.Q. Nguyen , Thinh G. Hoang , Linh H. Do A , Xuan T. Ngo ,
Phuong H.T. Nguyen , Giang D.L. Nguyen , Giang N.T. Nguyen.

Crowdfunding has transformed the way ventures secure funding, offering a dynamic platform where founders can seek support for a wide range of initiatives, be it for-profit ventures or projects with social or cultural impact. This method utilizes the power of internet and social media platforms to connect entrepreneurs with potential investors, enabling them to raise capital in exchange for future products or a stake in the venture. Technology, particularly blockchain, has emerged as a cornerstone in the crowdfunding landscape. Blockchain technology, known for its transparency and security, plays a pivotal role in fostering trust between investors and stakeholders. By leveraging blockchain, crowdfunding platforms can ensure that transactions are transparent and immutable, thus enhancing the credibility of the projects showcased on their platforms. This increased credibility can lead to greater success for ventures seeking funding, as investors are more inclined to contribute to projects that are backed by a reliable and transparent system.

TECHNIQUES: Tokenization, leveraging ,Smart Contracts, Transparency, Cryptocurrency,Ethereum.

MERITS: The content dives into technical details like React integration, MetaMask functionality, and Solidity coding, demonstrating a thorough understanding of the subject matter.

**[6] TITLE: CROWD FUNDING PLATFORMS USING BLOCKCHAIN
SMART CONTRACTS (2020).**

AUTHOR: Aakarshit Giri , Dr. Avinash Gupta ,Miss. Alka Singh , Atul Kumar Varma , Anchal Jha.

The Blockchain is a decentralized digital ledger that keeps track of all transactions. Because it is a distributed system, every node in the decentralized network stores all of the records. Smart Contracts are Ethereum-based programs that operate on the blockchain. The Ethereum Virtual Machine is used to operate all Smart Contracts. Crowdfunding makes it simple to raise funds for new project ideas. The issue with today's crowdfunding organizations is that they charge exorbitant fees, and fraud does occur sometimes. Avoiding these issues can be as simple as using a blockchain-based crowdfunding campaign. Traditional transaction costs and platform fees generally associated with other crowdfunding sites, such as Kickstarter, are eliminated by using Peer to Peer smart contracts for crowdfunding. Our project's goal is to develop a dependable application that will allow every fresh concept to come to reality. We created a crowdfunding platform that is built on the blockchain. In this program, we give an easy-to-use interface for anyone to generate and share their ideas. These ideas are then made available to the whole audience. Anyone who wants to support their ideals is welcome to do so. All of these procedures are carried out in a collaborative manner.

TECHNIQUES: Blockchain , Crowdfunding , Smart Contracts , Request – Approval.

MERITS: This platforms offer secure transactions, precise investment insights, global reach, and eliminate excessive fees associated with traditional platforms.

CHAPTER 3

SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

Smart contracts are self-executing contracts with the terms of the agreement directly written into code. These contracts automatically enforce and execute the terms of the agreement when predefined conditions are met. By utilizing smart contracts, the system ensures transparency in transactions because the terms of the contract are visible to all parties involved. This transparency helps to build trust among participants as they can verify the terms and conditions of the agreement without relying on intermediaries like banks or legal entities.

Cryptographic algorithms are mathematical functions used to secure and protect data by converting it into a format that is unreadable to unauthorized users. These algorithms play a crucial role in ensuring the confidentiality, integrity, and authenticity of data. By employing cutting-edge cryptographic algorithms, the system enhances data security by making it extremely difficult for unauthorized parties to access or manipulate sensitive information. This ensures that data transmitted within the system remains confidential and tamper-proof, mitigating the risk of data breaches or cyber-attacks.

Transaction fees are charges levied on users for processing their transactions within a blockchain network. These fees help incentivize miners or validators to include transactions in the blockchain and secure the network. However, if not managed effectively, high transaction fees can become a barrier to widespread adoption and usage of the system.

3.1.1 Disadvantages

- **Less Transparency:** While blockchain technology inherently provides transparency, the complexity of blockchain systems and smart contracts can sometimes make it challenging for users to understand the mechanics fully
- **Data Safety Concerns:** Blockchain technology itself is secure due to its decentralized and immutable nature. However, vulnerabilities in smart contracts or platform implementations can still pose risks to users' data safety.
- **High Transaction Fees:** Transactions on blockchain networks often incur fees, known as gas fees in Ethereum or transaction fees in Bitcoin. These fees can vary based on network congestion and the complexity of transactions.
- **Limited Regulation and Legal Frameworks:** The regulatory landscape surrounding blockchain and crowdfunding platforms is still evolving in many jurisdictions.
- **Volatility and Risk:** Cryptocurrencies, which are often used for transactions on blockchain-based crowdfunding platforms, are known for their price volatility. This volatility introduces an additional layer of risk for both project creators and backers.
- **Scalability Challenges:** Some blockchain networks face scalability limitations, especially during periods of high demand. This can lead to congestion and slower transaction processing times, affecting the user experience on crowdfunding platforms.

3.2 Problem Statement

Crowdfunding is a popular method for raising funds for various projects, causes, or individuals in need. However, traditional crowdfunding platforms often involve intermediaries and face challenges such as lack of transparency, high fees, and limited accessibility. This creates barriers for both fundraisers and backers, hindering the efficiency and effectiveness of the crowdfunding process . Integrating

blockchain technology into crowdfunding systems presents an opportunity to address these challenges by enabling direct peer-to-peer transactions, eliminating intermediaries, reducing costs, and providing greater transparency and security . Despite its potential benefits, implementing blockchain-based crowdfunding solutions requires overcoming technical, regulatory, and adoption hurdles. Therefore, there is a need to explore and develop innovative blockchain-based crowdfunding systems that can enhance accessibility, transparency, and efficiency while mitigating the associated challenges.

- **Transparency Deficiency:** Current crowdfunding platforms lack transparency, making them susceptible to fraudulent activities .
- **Fraudulent Activities:** The absence of transparent mechanisms in crowdfunding leads to fraudulent practices, undermining trust among participants .
- **Reliability Concerns:** Due to the lack of transparency, crowdfunding campaigns often fail to attract reliable investments, hindering project success .
- **Smart Contract Integration:** Implementing smart contracts requires addressing technical complexities and ensuring compatibility with existing crowdfunding platforms .

Objectives

- Develop a secure and transparent crowdfunding platform using Blockchain technology and smart contracts.
- Enhance trust among participants by providing visibility into campaign progress and ensuring tamper-proof transactions.
- Overcome technical challenges to seamlessly integrate smart contracts into existing crowdfunding platforms.
- Ensure regulatory compliance to operate within legal frameworks while leveraging Blockchain for crowdfunding operations.

3.3 PROPOSED SYSTEM

Ethereum is a blockchain platform that enables developers to create decentralized applications (dApps) and smart contracts. Smart contracts are self-executing contracts with the terms of the agreement directly written into code. They automatically enforce and execute the terms of the agreement when predetermined conditions are met, without the need for intermediaries. By leveraging Ethereum smart contracts, transactions can be made transparent and secure. Transparency is achieved because all transactions and smart contract interactions are recorded on the Ethereum blockchain, which is publicly accessible and immutable. Security is enhanced through cryptographic techniques, ensuring that transactions are tamper-proof and resistant to unauthorized access or alteration.

Encryption is the process of encoding information in such a way that only authorized parties can access it. By incorporating encryption techniques into the smart contract and communication channels, sensitive data can be protected from unauthorized access. MFA adds an extra layer of security by requiring users to provide multiple forms of verification before gaining access to a system or completing a transaction. Implementing MFA within Ethereum applications enhances security by reducing the risk of unauthorized access, even if one authentication factor is compromised.

Decentralized storage refers to the distribution of data across multiple nodes in a network rather than storing it on a centralized server. By utilizing decentralized storage solutions, such as IPFS (InterPlanetary File System) or Swarm, Ethereum applications can ensure data redundancy, availability, and censorship resistance. Immutable The Ethereum blockchain maintains an immutable ledger of transactions and smart contract states. Once data is recorded on the blockchain, it cannot be altered or deleted, ensuring the integrity and trustworthiness of records.

3.3.1 Advantages

- **Transparency:** Blockchain provides an immutable ledger, ensuring transparency in transactions. Every transaction is recorded on the blockchain, making it easier for backers to track where their funds are going.
- **Security:** Blockchain's decentralized nature makes it resistant to hacking and fraud. Smart contracts can automate the release of funds once certain conditions are met, reducing the risk of funds being misappropriated.
- **Lower Costs:** Traditional crowdfunding platforms often charge high fees for their services. By eliminating intermediaries, blockchain crowdfunding can significantly reduce transaction costs, allowing more funds to reach project creators.
- **Global Access:** Blockchain enables crowdfunding campaigns to reach a global audience without the need for traditional banking infrastructure. This opens up opportunities for creators to access a larger pool of potential backers.
- **Tokenization:** Blockchain allows for the creation of digital tokens representing ownership or rights to a project. These tokens can be traded on secondary markets, providing liquidity to backers and potentially increasing the value of their investment.
- **Increased Trust:** The transparent and immutable nature of blockchain transactions helps build trust between project creators and backers. Backers can verify the legitimacy of a project and its funding history on the blockchain.
- **Decentralization:** Unlike traditional crowdfunding platforms, which are often centralized and subject to censorship or regulatory issues, blockchain crowdfunding platforms operate in a decentralized manner, reducing the risk of interference or shutdowns.

CHAPTER 4

SYSTEM DESCRIPTION

4.1.SYSTEM SPECIFICATION

4.1.1 Hardware Requirements

- **Processor** : Intel Core 5 processor
- **RAM** : 4 GB
- **Hard disk** :160 GB
- **Keyboard** : Standard Keyboard.
- **Monitor** : 15 inch color monitor

4.1.2 Software Requirements

- **Operating system** : Windows 10
- **Language** : Shell, Solidity
- **Tools** : Solidity, Next.js, web3.js
- **Styling** : CSS, JavaScript

Programming Language: Solidity

Solidity stands out as a dedicated programming language tailored for crafting smart contracts on blockchain platforms such as Ethereum. Within the crowdfunding realm, Solidity finds application in crafting smart contracts that oversee the entire crowdfunding process. These contracts serve as the backbone, outlining precise regulations for accepting contributions, disbursing funds to project creators, and managing various contingencies or occurrences pertinent to the crowdfunding campaign. Through its syntax and functionalities, Solidity facilitates the creation of robust and transparent frameworks, ensuring the seamless execution of crowdfunding operations while upholding the principles of decentralization and trust within the blockchain ecosystem.

Next.js is renowned as a React framework, is widely embraced for constructing web applications with unparalleled efficiency. Its versatility extends seamlessly to the realm of crowdfunding platforms, empowering developers to craft captivating front-end interfaces. Leveraging the capabilities of Next.js, developers can fabricate dynamic and interactive user experiences that seamlessly interact with the Ethereum blockchain through web3.js integration. This amalgamation facilitates real-time data fetching, smart contract interaction, and seamless user interactions, thus enriching the crowdfunding experience with enhanced transparency and user engagement.

web3.js serves as a pivotal JavaScript library facilitating seamless interaction with Ethereum nodes, thereby bridging the gap between your web application and the Ethereum blockchain. This essential tool empowers users to engage directly with smart contracts deployed on Ethereum, unlocking a realm of possibilities within the context of crowdfunding.

CHAPTER 5

SYSTEM DESIGN

Front-end

For the frontend, developers typically utilize standard web technologies like HTML, CSS, and JavaScript to create a user-friendly interface. This interface interacts with the blockchain via client-side libraries such as Web3.js or Ether.js, facilitating seamless communication between users and the underlying smart contracts.

Back-end

The backend comprises the smart contract code deployed on the blockchain, responsible for managing the crowdfunding process. Developers write and deploy these smart contracts, defining rules for contributions, fund distribution, and other platform functionalities.

Database

In terms of the database, traditional relational databases may not be necessary since blockchain technology inherently provides an immutable and decentralized ledger. Smart contracts store data on the blockchain, ensuring transparency and trustworthiness without the need for a centralized database.

Integration

Integration with external services or protocols may be necessary to enhance the platform's functionality. For example, integrating with payment gateways or identity verification services can streamline the crowdfunding process and enhance user experience. Additionally, interoperability with other blockchain protocols or decentralized finance (DeFi) platforms can unlock new opportunities for fundraising and investment.

5.1 ARCHITECTURE DIAGRAM

Smart contracts are simply programs stored on a blockchain that run when predetermined conditions are met. They typically are used to automate the execution of an agreement so that all participants can be immediately certain of the outcome, without any intermediary's involvement or time loss.

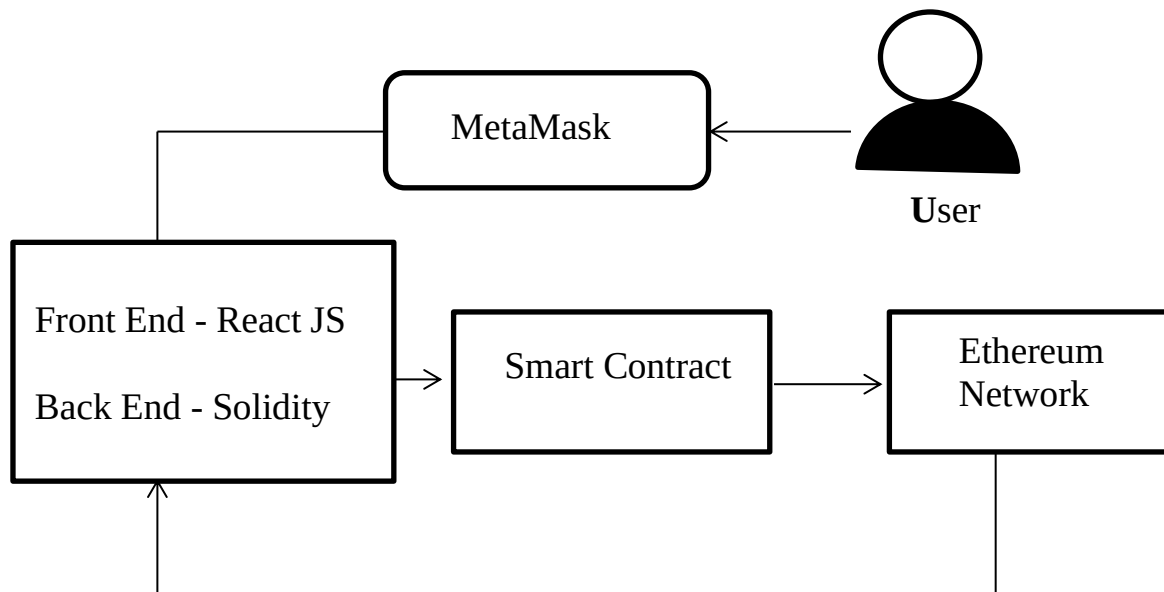


Fig.No: 5.1 Architecture Diagram

The system architecture of this project can be divided into three main components:

Frontend (website)

The frontend or client side of a website can use standard web technologies such as HTML and JavaScript. This allows developers to use familiar tools, libraries, and frameworks. The client-side user interface is usually linked to the smart contracts via client-side libraries such as Web3.js or Ether.js which are bundled with the frontend resources and sent to the browser together with the UI. Interactions with smart contracts such as signing messages and sending transactions to the smart contracts are usually conducted via a browser-based Web3 wallet, such as MetaMask.

Backend (Server)

The backend component is responsible for managing the crowdfunding process. Developers write and deploy these smart contracts, defining rules for contributions, fund distribution, and other platform functionalities. The backend component consists of three main modules:

Creating a Campaign

When a user initiates a campaign on the crowdfunding platform, they embark on a journey to share their project or cause with the world. Through a straightforward process, they furnish essential details such as the project's description, its funding objective, and the envisioned timeline for the campaign's duration. Additionally, they may offer incentives or rewards to entice potential contributors, enriching the crowdfunding experience.

Contributing a Campaign

Once a campaign is live on the crowdfunding platform, it becomes accessible to potential supporters eager to contribute to the cause. Interested users navigate to the platform, where they can peruse the diverse array of campaigns available. Upon finding a project that resonates with them, they seamlessly initiate the contribution process by selecting the campaign of their choice.

Withdrawal of Funds

The campaign initiator possesses the privilege to kickstart the funds withdrawal process. Harnessing the automated functionalities of the smart contract, the withdrawal unfolds effortlessly, abiding by the predetermined regulations ingrained within the contract. This seamless procedure ensures transparency and trust, maintaining the integrity of the crowdfunding ecosystem.

5.2 USE CASE DIAGRAM

The Use Case Diagram for the Crowdfunding platform using blockchain includes two primary actors - the user and the investor. The user can do register project, fund for project. The diagram highlights the main functionalities of the system and how they are used by the actors.

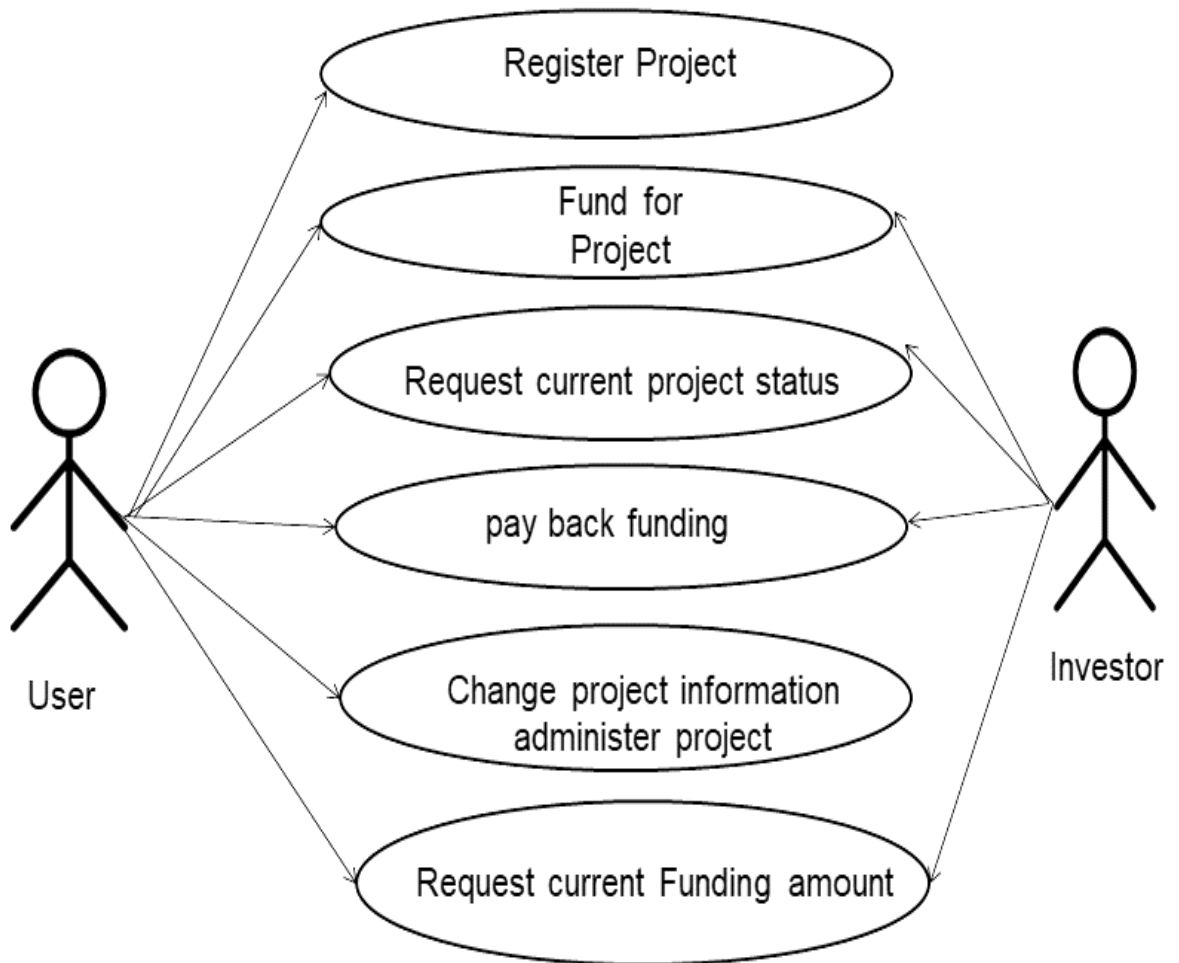


Fig.No: 5.2 Use Case Diagram

GOALS

Crowdfunding utilizing blockchain technology is driven by several key goals aimed at transforming the conventional fundraising landscape. One of the paramount objectives is to introduce an unprecedented level of transparency into the crowdfunding process. Unlike traditional crowdfunding platforms where transactions are often opaque and subject to manipulation, blockchain offers an immutable ledger that records every contribution and transaction in a transparent and tamper-proof manner. This transparency fosters trust among participants, as contributors can verify where their funds are going and how they are being utilized.

Security is another critical goal of blockchain-based crowdfunding. By leveraging cryptographic techniques and decentralized consensus mechanisms, blockchain ensures that transactions are secure and resistant to hacking or fraudulent activities. Smart contracts, which are self-executing contracts with the terms of the agreement directly written into code, further enhance security by automating the release of funds based on predefined conditions. This eliminates the need for intermediaries such as banks or payment processors, reducing the risk of funds being misappropriated or delayed.

Efficiency is also a key goal of crowdfunding using blockchain. Traditional crowdfunding platforms often involve lengthy and cumbersome processes for verifying transactions, disbursing funds, and resolving disputes. With blockchain, these processes can be streamlined and automated, leading to faster and more cost-effective fundraising campaigns. Additionally, blockchain enables crowdfunding on a global scale, allowing creators to reach a broader audience and contributors to participate in campaigns from anywhere in the world without the need for complex currency conversions or international transfer fees.

CHAPTER 6

SYSTEM IMPLEMENTATION

6.1 MODULES

In a crowdfunding platform built using blockchain technology, several key modules contribute to its functionality and security. The project typically encompasses modules for user registration and authentication, project creation and management, payment processing, smart contract execution, and transparency features.

- CREATING A CAMPAIGN
- CONTRIBUTING A CAMPAIGN
- WITHDRAWAL OF FUNDS

DIAGRAM

The diagram is represent a crowdfunding platform built on blockchain technology, several key modules are essential to facilitate the seamless operation of campaigns and fund management. The first module involves creating a campaign, where project creators can outline their goals, objectives, and funding targets. This module typically includes features such as setting fundraising milestones, defining campaign duration, and providing detailed descriptions and multimedia content to attract potential backers. The second module revolves around contributing to a campaign, allowing users to pledge funds towards projects they wish to support. This module facilitates secure transactions using blockchain technology, ensuring transparency, immutability, and tamper-proof record-keeping. Users may have the option to contribute using various cryptocurrencies or fiat currencies, with their contributions recorded on the blockchain for accountability and auditability.

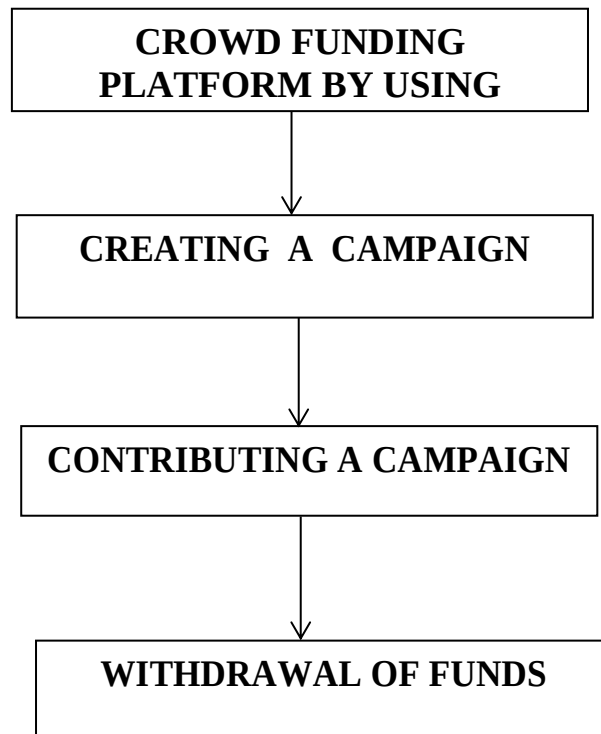


Fig.No: 6.1 Modules

The third module involves the withdrawal of funds, enabling project creators to access the funds raised once a campaign reaches its funding goal or deadline. Withdrawal processes are typically governed by predefined rules and smart contracts, which automatically release funds to the project creator's designated wallet upon meeting specified criteria. This module ensures that funds are disbursed efficiently and transparently, minimizing delays and potential disputes. Overall, these modules form the foundation of a blockchain-based crowdfunding platform, providing creators and backers with a transparent, secure, and decentralized environment for initiating, supporting, and managing crowdfunding campaigns. By leveraging blockchain technology, such platforms offer greater trust, accountability, and efficiency compared to traditional crowdfunding platforms, ultimately benefiting both project creators and backers alike.

6.2 MODULES DESCRIPTION

6.2.1 Creating a Campaign

Creating a campaign on a crowdfunding platform powered by blockchain technology, it begins with the initiation of a project or cause. This initiation involves detailing the purpose, goals, and the target funding amount required to achieve those goals. One of the significant advantages of utilizing blockchain in this process is the immutable storage of information. Every detail about the campaign, from its inception to its objectives, is recorded on the blockchain, creating a transparent and tamper-proof record. smart contracts play a pivotal role in automating various aspects of the campaign. For instance, they can be programmed to set fundraising milestones, ensuring that progress is tracked transparently and that contributors are aware of the campaign's advancement. Additionally, smart contracts can automate the release of funds when specific conditions are met, such as reaching a fundraising goal or completing a project milestone. This automation not only streamlines the fundraising process but also enhances trust among participants, as the terms and conditions are enforced automatically by the blockchain network.

6.2.2 Contributing a Campaign

Contributing to a campaign involves individuals or entities pledging funds to support the project or cause. This process is traditionally conducted through various means, such as online payment gateways or direct transfers. However, blockchain technology offers a revolutionary approach to this process by providing a decentralized, transparent, and secure platform for recording contributions.

Blockchain facilitates secure transactions by leveraging cryptographic techniques to ensure the integrity and confidentiality of the data exchanged. Each contribution made to a crowdfunding campaign is recorded as a transaction on the blockchain ledger, which is distributed across a network of nodes. This decentralized nature of blockchain eliminates the need for a central authority to oversee transactions, reducing the risk of fraud, manipulation, or unauthorized

alterations. Moreover, smart contracts play a pivotal role in governing contributions within a blockchain-based crowdfunding campaign. These self-executing contracts are programmed to enforce predefined rules and conditions, ensuring that contributions are made in accordance with the campaign's requirements. For instance, smart contracts can verify the identity of contributors by requiring them to fulfill specific authentication criteria, such as cryptographic signatures or biometric verification.

Furthermore, smart contracts can enforce contribution limits to prevent individuals from pledging more funds than allowed or to ensure fair distribution of rewards or incentives. These limits can be predefined within the smart contract code, and once reached, the contract automatically stops accepting additional contributions, thereby maintaining transparency and integrity in the fundraising process.

6.2.3 Withdrawal of Funds

Withdrawal of funds occurs when the campaign reaches its funding goal or specific milestones. Blockchain technology ensures that funds are disbursed according to predetermined conditions encoded in smart contracts, reducing the risk of fraud or mismanagement. Additionally, the decentralized nature of blockchain enhances trust by providing transparent visibility into how funds are allocated and utilized. This automation reduces administrative overhead and potential delays associated with traditional manual fund disbursement methods. Moreover, the immutable nature of blockchain transactions provides an auditable trail of fund withdrawals, enhancing accountability and transparency for all stakeholders involved. This transparent visibility into fund allocation and utilization instills confidence among contributors, fostering a greater sense of trust in the crowdfunding campaign's management and execution.

ALGORIHTM

Ethereum Smart Contract

Withdrawal of funds occurs when the campaign reaches its funding goal or specific milestones. A crowdfunding platform built on blockchain technology, particularly leveraging Ethereum's smart contract capabilities, offers a revolutionary approach to fundraising with numerous advantages. At its core, Ethereum's smart contracts enable the creation of self-executing agreements where the terms are directly coded into the blockchain. This automation ensures that transactions are executed exactly as agreed upon, without the need for intermediaries, thus fostering transparency, security, and efficiency. In the realm of crowdfunding, Ethereum smart contracts serve as the backbone of a decentralized fundraising mechanism. Contributors can participate in campaigns by sending cryptocurrency directly to the smart contract address. These contributions are securely recorded on the blockchain, eliminating the risk of fraud or manipulation often associated with centralized platforms. Importantly, funds are only released to the project creators once predetermined conditions, such as reaching a funding goal, are met. This feature safeguards the interests of backers and incentivizes project creators to meet their objectives. Moreover, Ethereum smart contracts can enforce rules for project milestones, ensuring that funds are released incrementally as progress is made. This transparent tracking of fund allocation provides backers with visibility into how their contributions are being utilized, enhancing accountability and trust in the platform. By harnessing the power of blockchain technology and Ethereum smart contracts, crowdfunding platforms can revolutionize the fundraising process, empowering creators and backers alike. With greater transparency, security, and efficiency, these platforms enable participants to engage in innovative projects with confidence, driving forward a new era of decentralized fundraising.

6.3 IMPLEMENTATION

Withdrawal of funds occurs when the campaign reaches its funding goal or specific milestones. Blockchain technology ensures that funds are disbursed according to predetermined conditions encoded in smart contracts, reducing the risk of fraud or mismanagement. In the development of a crowdfunding platform utilizing blockchain technology, several key implementation aspects come into play. Firstly, blockchain's inherent transparency and immutability can be leveraged to provide a secure and transparent fundraising environment. Smart contracts, programmable self-executing contracts deployed on the blockchain, can automate various aspects of the crowdfunding process, such as fund distribution and project milestone verification, reducing the need for intermediaries and enhancing trust among participants. The platform can incorporate features such as multi-signature wallets to ensure that funds are released only upon consensus from both the project creators and backers, thereby mitigating the risk of fraudulent activities. Additionally, utilizing blockchain-based tokens for contributions can enable fractional ownership and facilitate the trading of project tokens on secondary markets, providing liquidity to backers and potentially increasing the overall funding capacity of the platform. Implementing a user-friendly interface and providing educational resources on blockchain technology can also help onboard non-technical users and foster wider adoption of the platform. Furthermore, leveraging blockchain's traceability can enable donors to track the flow of funds and ensure that their contributions are used for their intended purposes, fostering greater accountability and transparency within the crowdfunding ecosystem. Overall, by harnessing the capabilities of blockchain technology, a crowdfunding platform can offer a more efficient, transparent, and inclusive fundraising experience for both project creators and backers.

CHAPTER 7

SYSTEM TESTING

7.1 TESTING OBJECTIVES

The objectives for crowdfunding platforms leveraging blockchain technology are multifaceted. Firstly, ensuring transparency and accountability in fundraising activities is paramount. By utilizing blockchain's immutable ledger, every transaction, from donations to project updates, can be recorded transparently, enhancing trust among backers. Secondly, security is a crucial concern. Blockchain's cryptographic features offer robust protection against fraud and data tampering, safeguarding both contributors' investments and project creators' integrity. Additionally, decentralization facilitates global participation, enabling crowdfunding campaigns to reach a broader audience without the need for intermediaries or restrictive regulations. Furthermore, smart contracts automate the execution of crowdfunding agreements, streamlining processes and reducing administrative overhead. Overall, the primary objectives encompass transparency, security, decentralization, and efficiency, ultimately fostering a more inclusive and trustworthy crowdfunding ecosystem.

7.2 TYPES OF TESTING

System Testing

System testing is testing for crowdfunding platforms that utilize blockchain technology requires a comprehensive approach to ensure the security, integrity, and functionality of the system. Firstly, testing must verify the smart contracts governing transactions to guarantee their accuracy and resistance to vulnerabilities such as transparency or overflow attacks.

Unit Testing

Unit testing for crowdfunding platforms using blockchain involves testing individual components or units of the platform's codebase to ensure they function

correctly in isolation. This testing approach is crucial for verifying the functionality of smart contracts, transaction processing, and data integrity within the blockchain network.

Functional Testing

Functional testing for crowdfunding platforms that utilize blockchain technology involves verifying various aspects of the platform's functionality to ensure smooth operation and user satisfaction. Test scenarios should cover different user roles, including project creators, backers, and administrators, to ensure that each user can interact with the platform effectively.

Integration Testing

The integration testing process involves creating test cases that simulate different usage scenarios to ensure that the modules work correctly when integrated. The tests are performed manually or through automated testing tools to ensure that the system functions as intended.

White Box Testing

White box testing involves creating test cases based on the system's internal design and structure. This testing method ensures that the system's internal structure and design are sound and that the system functions correctly as intended.

Black Box Testing

Black box testing for a blockchain-based crowdfunding project, the primary focus is on assessing the platform's functionality from an end-user perspective, without delving into its internal workings. Testers approach this evaluation by creating diverse scenarios that mimic real-world interactions with the platform, such as user registration, project creation, donation processing, and reward distribution.

7.3 TEST CASES

For testing a crowdfunding platform that utilizes blockchain technology, It want to consider various scenarios to ensure the platform works correctly and securely. Here's a basic outline of some test cases might consider:

1. User Registration

- Test user registration process, ensuring all required fields are validated.
- Verify that each user gets a unique identifier or address on the blockchain.

2. Project Creation

- Test creating a new project, ensuring all required details are entered.
- Verify that the project is correctly recorded on the blockchain.

3. Contributing to a Project

- Test contributing to a project, ensuring the transaction is recorded on the blockchain.
- Verify that the contributed funds are correctly reflected in the project's total funding.

4. Project Updates

- Test updating a project's details, such as the funding goal or project description.
- Verify that the updates are correctly recorded on the blockchain and displayed to users.

5. Project Completion

- Test completing a project, ensuring that any remaining funds are distributed correctly.
- Verify that the project status is updated on the blockchain.

6. Security Testing

- Test for vulnerabilities such as SQL injection, cross-site scripting (XSS), and other common web security issues.
- Verify that transactions are secure and cannot be tampered with.

7. Performance Testing

- Test the platform's performance under various load conditions to ensure it can handle multiple users and transactions simultaneously.

8. Blockchain Integration Testing

- Test the integration with the blockchain network to ensure transactions are being recorded correctly and in a timely manner.

- Verify that the platform interacts correctly with the blockchain's smart contracts.

9. User Interface Testing

- Test the user interface to ensure it is intuitive and easy to use.
- Verify that users can easily navigate the platform and perform actions such as contributing to projects.

10. Compatibility Testing

- Test the platform on different browsers and devices to ensure compatibility.
- Verify that the platform works correctly on mobile devices and tablets.

Result Output:

Our blockchain-based crowdfunding platform revolutionizes project funding by ensuring decentralized transactions and transparency through immutable ledger records. With smart contracts, funds are released to creators only upon achieving predefined milestones, mitigating risks for backers. Global accessibility eliminates geographical barriers, enabling participation from anyone, anywhere. Tokenization facilitates investment in projects worldwide, fostering a diverse ecosystem of innovation and collaboration. By leveraging blockchain technology, we aim to democratize funding and empower creators to bring their ideas to life. Together, we can create a more inclusive and transparent future for project funding.

CHAPTER 8

RESULTS AND DISCUSSION

8.1.PERFORMANCE ANALYSIS

In a crowdfunding platform leveraging blockchain technology, performance analysis is crucial for assessing the platform's effectiveness, transparency, and trustworthiness. Blockchain's distributed ledger ensures that all transactions and funding activities are transparent and immutable, providing a reliable record for performance evaluation. Smart contracts can automate many aspects of the crowdfunding process, such as fund allocation and distribution, enhancing efficiency and reducing the risk of fraud. Additionally, blockchain can enable global participation in crowdfunding campaigns, opening up new avenues for fundraising and investment. Performance analysis in this context would focus on metrics such as the number of successful campaigns, funds raised, user satisfaction, and the platform's ability to facilitate secure and transparent transactions. Overall, a crowdfunding platform built on blockchain can offer a secure, transparent, and efficient way to raise funds, with performance analysis playing a critical role in ensuring its success and sustainability.

Accuracy

Accuracy in a crowdfunding platform powered by blockchain is akin to the bedrock of trust and reliability upon which the entire fundraising endeavor rests. The immutable ledger characteristic of blockchain technology stands as an unassailable fortress, ensuring that every transaction, contribution, and disbursement is etched into history with absolute precision, impervious to any attempts at alteration or manipulation. This transparent and unalterable record serves as a beacon of trust, offering stakeholders a clear window into the platform's operations and fostering an environment of integrity. Smart contracts, the digital guardians of predefined rules and conditions, further fortify accuracy by automating critical processes like fund allocation and distribution, leaving no

room for ambiguity or error. Moreover, the decentralized architecture of blockchain acts as a distributed network of guardians, each node meticulously validating transactions to maintain the sanctity of the ledger. With accuracy at its core, stakeholders find solace in the reliability of the crowdfunding platform, nurturing a culture of trust and confidence that fuels greater participation and investment, ultimately propelling the project towards success and sustainability.

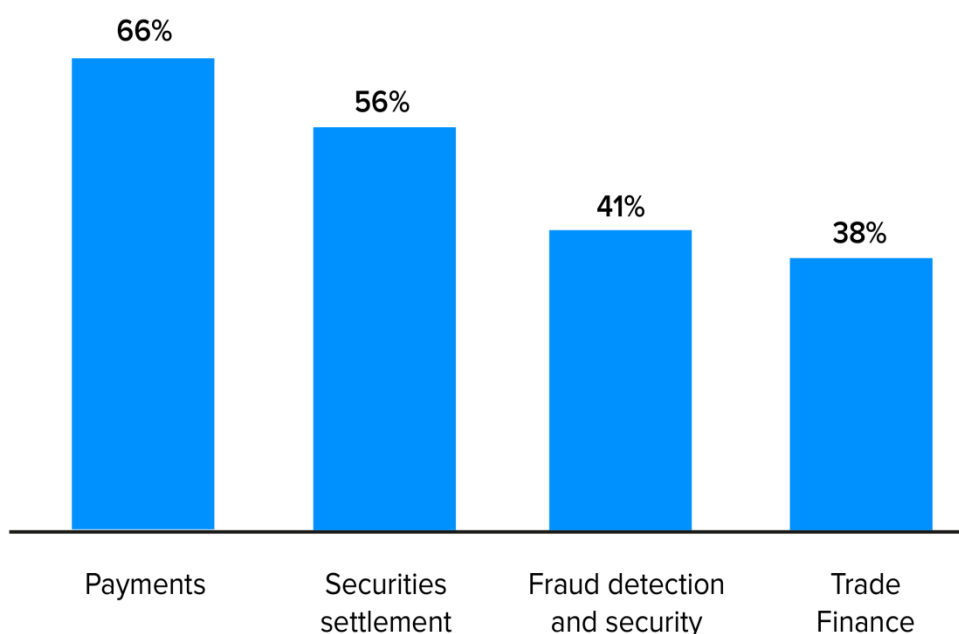


Fig.No: 8.1 Classification Report Analysis

In a crowdfunding platform utilizing blockchain technology, accuracy is paramount to ensuring the integrity and reliability of the fundraising process. Blockchain's immutable ledger ensures that all transactions, contributions, and disbursements are recorded accurately and cannot be tampered with, providing a transparent and trustworthy record of all activities. Smart contracts, programmed to execute predefined actions when specific conditions are met, further enhance accuracy by automating various aspects of the crowdfunding process, such as fund allocation and distribution. Additionally, blockchain's decentralized nature minimizes the risk of errors or manipulation, as transactions are validated by multiple nodes across the network. With accurate and transparent record-keeping facilitated by blockchain, stakeholders can have confidence in the reliability of the

crowdfunding platform, fostering trust among participants and encouraging greater participation and investment. In the context of a crowdfunding platform utilizing blockchain technology, an equation commonly used is the cryptographic hashing algorithm. One example is the Keccak algorithm, which is used to create a unique cryptographic hash value for a given input data. The equation for the Keccak algorithm involves multiple rounds of permutation and substitution operations, resulting in a fixed-size output that uniquely represents the input data. This equation can be represented symbolically as:

$$H = \text{Keccak}(M)$$

Where:

- H represents the resulting hash value.
- M represents the input message or data.

Implementing this equation within the project involves applying the Keccak algorithm to critical data such as transaction details, user identities, and smart contract parameters, ensuring their security and integrity on the blockchain ledger.

8.2 TRAINING ANALYSIS

In a crowdfunding platform powered by blockchain technology, training analysis plays a pivotal role in ensuring the platform's efficiency, security, and effectiveness. Blockchain's distributed ledger system necessitates thorough training of personnel responsible for managing and overseeing the platform's operations.

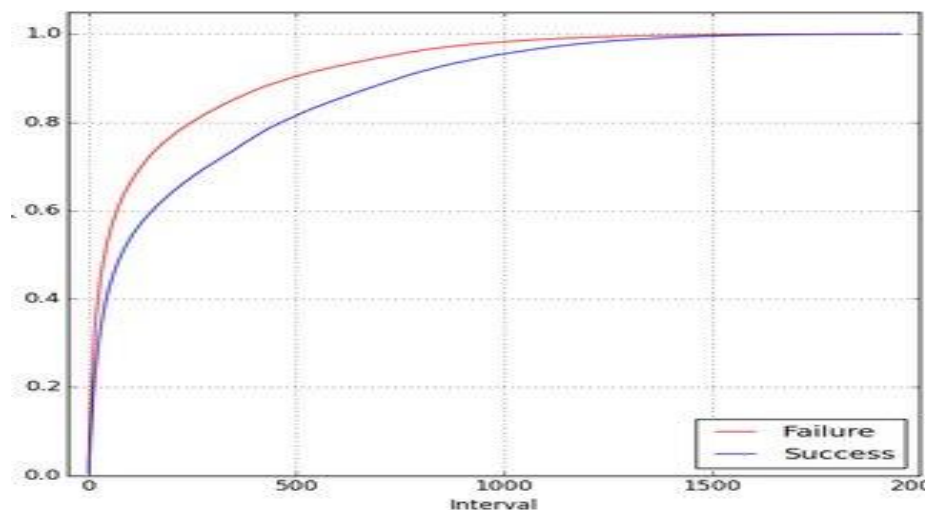


Fig.No: 8.2 Training Analysis

Training encompasses understanding the intricacies of blockchain technology, including its cryptographic principles, consensus mechanisms, and smart contract functionalities. Additionally, training analysis involves educating personnel on best practices for managing crowdfunding campaigns, handling digital assets securely, and addressing potential security threats such as hacking attempts or fraudulent activities. Furthermore, continuous training and analysis are essential to keep up with evolving blockchain technologies and emerging cybersecurity challenges. By investing in comprehensive training and analysis initiatives, crowdfunding platforms can enhance their operational capabilities, mitigate risks, and maintain the trust and confidence of their users.

In a crowdfunding platform driven by blockchain technology, training analysis holds a crucial position in ensuring smooth operations, robust security, and optimal performance. Given the decentralized nature of blockchain, personnel overseeing the platform must undergo extensive training to grasp the intricacies of this innovative technology. This training encompasses a deep understanding of blockchain's cryptographic principles, consensus mechanisms, and the functionalities of smart contracts. Moreover, training analysis involves imparting knowledge on the best practices for effectively managing crowdfunding campaigns, securely handling digital assets, and swiftly addressing potential security threats like hacking attempts or fraudulent activities. Continuous training and analysis are imperative to staying abreast of evolving blockchain technologies and emerging cybersecurity challenges. By prioritizing comprehensive training and analysis initiatives, crowdfunding platforms can fortify their operational capabilities, mitigate risks, and uphold the trust and confidence of their users, thereby fostering a robust and sustainable crowdfunding ecosystem.

CHAPTER 9

APPENDICES

9.1 SAMPLE CODE

```
// SPDX-License-Identifier: UNLICENSED
```

```
pragma solidity ^0.8.9;
```

```
contract CANFund
```

```
{
```

```
    event FundCreated(
```

```
        uint256 indexed fundId,
```

```
        address indexed proprietor,
```

```
        string CANname,
```

```
        string about,
```

```
        uint256 fundNeeded,
```

```
        uint256 timeCutOff,
```

```
        string poster
```

```
    );
```

```
    struct CAN
```

```
    {
```

```
        address proprietor;
```

```
        string CANname;
```

```
        string about;
```

```
        uint256 fundNeeded;
```

```
        uint256 timeCutOff;
```

```
        uint256 fundCollected;
```

```
        string poster;
```

```
        address[] benefactors;
```

```
        uint256[] endowments;
```

```
    }
```



```

mapping(uint256 => CAN) public funds;
mapping(address => uint256) public userCANBalance;
mapping(address => bool) public authenticatedUsers;
uint256 public totalCANs = 0;
function authenticateUser(bytes32 message, uint8 v, bytes32 r, bytes32 s) public
view returns (bool)
{
    address signatory = ecrecover(message, v, r, s);
    return signer == msg.sender;
}

```

```

function createCAN(address _proprietor, string memory _CANname, string
memory _about, uint256 _fundNeeded, uint256 _timeCutOff, string memory
_poster) public returns (uint256)
{
    CAN storage fund = funds[totalCANs];
    require(fund.timeCutOff < block.timestamp, "Invalid Time Cut Off!");
    fund.proprietor = _proprietor;
    fund.CANname = _CANname;
    fund.about = _about;
    fund.fundNeeded = _fundNeeded;
    fund.timeCutOff = _timeCutOff;
    fund.fundCollected = 0;
    fund.poster = _poster;
    totalCANs++;
    emit FundCreated(totalCANs - 1, _proprietor, _CANname, _about,
_fundNeeded, _timeCutOff, _poster);

    return (totalCANs - 1);
}

```

```

function getCANs() public view returns (CAN[] memory)

```

```

{
    CAN[] memory totalFunds = new CAN[](numberOfCampaigns);
    uint i = 0;

    for(; i < numberOfCampaigns; i++)
    {
        CAN storage tempFund = funds[i];
        totalFunds[i] = tempFund;
    }

    return totalFunds;
}

function endowCAN(uint256 _id, bytes32 message, uint8 v, bytes32 r, bytes32 s)
public payable
{
    require(authenticateUser(message, v, r, s), "Invalid signature");
    uint256 fundsEndowing = msg.value;
    CAN storage fund = funds[_id];
    fund.benefactors.push(msg.sender);
    fund.endowments.push(fundsEndowing);
    uint256 canCredit = ((fundsEndowing * 2) / 100);
    userCANBalance[msg.sender] += canCredit;
    (bool sent,) = payable(fund.proprietor).call{value: fundsEndowing}("");
    if(sent)
    {
        fund.fundCollected = fund.fundCollected + fundsEndowing;
    }
}

function getBenefactors(uint256 _id) view public returns (address[] memory,
uint256[] memory)
{
    return (funds[_id].benefactors, funds[_id].endowments);
}

```

```

}
import React from 'react'

const CountBox = ({ title, value }) => {
  return (
    <div className="flex flex-col items-center w-[150px]">
      <h4 className="font-epilogue font-bold text-[30px] text-white p-3 bg-yellow-400
rounded-t-[10px] w-full text-center truncate">{value}</h4>
      <p className="font-epilogue font-normal text-[16px] text-[#808191] bg-green-400
px-3 py-2 w-full rounded-b-[10px] text-center">{title}</p>
    </div>
  )
}

export default CountBox

import React from 'react'

const CustomButton = ({ btnType, title, handleClick, styles }) => {
  return (
    <button
      type={btnType}
      className={`font-epilogue font-semibold text-[16px] leading-[26px] text-white
min-h-[52px] px-4 rounded-[10px] ${styles}`}
      onClick={handleClick}
    >
      {title}
    </button>
  )
}

export default CustomButton

import React from 'react';
import { useNavigate } from 'react-router-dom';
import { v4 as uuidv4 } from 'uuid';
import FundCard from './FundCard';
import { loader } from '../assets';

```

```

const DisplayCampaigns = ({ title, isLoading, campaigns }) => {
  const navigate = useNavigate();

  const handleNavigate = (campaign) => {
    navigate(`/campaign-details/${campaign.title}`, { state: campaign });
  };

  const staticCampaigns = [
    {
      owner: 'pqr',
      title: 'Healthcare Outreach Program',
      description: 'Providing medical care and vaccinations to remote areas',
      amountCollected: 600,
      target: 800,
      deadline: 5,
      image:
        'https://image.freepik.com/free-vector/healthcare-outreach-program_23-2148561232.jpg',
    }
  ]
  return (
    <div>
      <h1 className="font-epilogue font-semibold text-[18px] text-white text-left">
        {title} ({campaigns.length + staticCampaigns.length})
      </h1>
      <div className="flex flex-wrap mt-[20px] gap-[26px]">
        {isLoading && <img src={loader} alt="loader" className="w-[100px] h-[100px] object-contain" />}

        {!isLoading && campaigns.length === 0 && staticCampaigns.length === 0
        && (
          <p className="font-epilogue font-semibold text-[14px] leading-[30px] text-

```

```

    [#818183]">
      You have not created any campaigns yet.
    </p>
  )}
  {!isLoading &&
    campaigns.map((campaign) => (
      <FundCard
        key={uuidv4()}
        {...campaign}
        handleClick={() => handleNavigate(campaign)}
      />
    ))}

</div>
</div>
);
};
export default DisplayCampaigns;
import React from 'react'

const FormField = ({ labelName, placeholder, inputType, isTextArea, value,
handleChange }) => {
  return (
    <label className="flex-1 w-full flex flex-col">
      {labelName && (
        <span className="font-epilogue font-medium text-[14px] leading-[22px] text-
[#808191] mb-[10px]">{labelName}</span>
      )}
      {isTextArea ? (
        <textarea
          required
          value={value}
          onChange={handleChange}

```

```

      rows={10}
      placeholder={placeholder}
      className="py-[15px] sm:px-[25px] px-[15px] outline-none border-[1px]
border-[#3a3a43] bg-transparent font-epilogue text-white text-[14px]
placeholder:text-[#4b5264] rounded-[10px] sm:min-w-[300px]"
    />
  ) : (
    <input
      required
      value={value}
      onChange={handleChange}
      type={inputType}
      step="0.1"
      placeholder={placeholder}
      className="py-[15px] sm:px-[25px] px-[15px] outline-none border-[1px]
border-[#3a3a43] bg-transparent font-epilogue text-white text-[14px]
placeholder:text-[#4b5264] rounded-[10px] sm:min-w-[300px]"
    />
  )}
</label>
)
}

```

```

import React from 'react';
import { tagType, thirdweb } from '../assets';
const FundCard = ({ owner, title, description, target, deadline, amountCollected,
image, handleClick }) => {

  return (
    <div className="sm:w-[288px] w-full rounded-[15px] bg-[#1c1c24] cursor-
pointer" onClick={handleClick}>
      <img src={image} alt="fund" className="w-full h-[158px] object-cover
rounded-[15px]" />

```

```

<div className="flex flex-col p-4">
  <div className="flex flex-row items-center mb-[18px]">
    <img src={tagType} alt="tag" className="w-[17px] h-[17px] object-contain"
    />
    <p className="ml-[12px] mt-[2px] font-medium text-[12px] text-
[#808191]">Charity</p>
  </div>

  <div className="block">
    <h3 className="font-semibold text-[16px] text-white text-left leading-[26px]
truncate">{title}</h3>
    <p className="mt-[5px] font-normal text-[#808191] text-left leading-[18px]
truncate">{description}</p>
  </div>

  <div className="flex justify-between flex-wrap mt-[15px] gap-2">
    <div className="flex flex-col">
      <h4 className="font-semibold text-[14px] text-[#b2b3bd] leading-
[22px]">{amountCollected}</h4>
      <p className="mt-[3px] font-normal text-[12px] leading-[18px] text-
[#808191] sm:max-w-[120px] truncate">Raised of {target}</p>
    </div>
    <div className="flex flex-col">
      <h4 className="font-semibold text-[14px] text-[#b2b3bd] leading-
[22px]">{deadline}</h4>
      <p className="mt-[3px] font-normal text-[12px] leading-[18px] text-
[#808191] sm:max-w-[120px] truncate">Days Left</p>
    </div>
  </div>

  <div className="flex items-center mt-[20px] gap-[12px]">
    <div className="w-[30px] h-[30px] rounded-full flex justify-center items-
center bg-[#13131a]">
      <img src={thirdweb} alt="user" className="w-1/2 h-1/2 object-contain" />
    </div>
  </div>

```

```

    </div>
    <p className="flex-1 font-normal text-[12px] text-[#808191] truncate">by
    <span className="text-[#b2b3bd]">{owner}</span></p>
  </div>
</div>
</div>
)
}
export default FundCard
import React from 'react';
import { loader } from '../assets';

const Loader = () => {
  return (
    <div className="fixed inset-0 z-10 h-screen bg-[rgba(0,0,0,0.7)] flex items-center justify-center">
      <div className="bg-white rounded-full p-6 shadow-md flex items-center space-x-4">
        <img src={loader} alt="loader" className="w-[50px] h-[50px] animate-spin" />
        <div>
          <p className="font-epilogue font-semibold text-lg text-gray-800">
            Transaction in Progress
          </p>
          <p className="text-gray-600">Please wait...</p>
        </div>
      </div>
    </div>
  );
};
export default Loader;
import React, { useEffect, useState } from "react";
import { Link, useNavigate } from "react-router-dom";
import { useContext } from "../context";
import { CustomButton } from ".";
import { sun } from "../assets";

```



```

const Navbar = () => {
  const navigate = useNavigate();
  const [isActive, setIsActive] = useState("dashboard");
  const [toggleDrawer, setToggleDrawer] = useState(false);

  const { connect, address } = useStateContext();

  return (
    <div className="flex md:flex-row flex-col-reverse justify-between mb-8 gap-6">
      <div className="absolute sm:flex hidden flex-row justify-end gap-4 right-5 top-5">

        <CustomButton
          btnType="button"
          title={address ? "Create CAN" : "Connect"}
          styles={address ? "bg-[#3498db]" : "bg-[#8c6dfd]"}
          handleClick={() => {
            if (address) {
              navigate("create-campaign");
            } else connect();
          }}
        />
      </div>
    </div>
  );
}

```

9.2. SCREENSHOTS

Creating a Campaign Page

Create New Campaign

Title

Description

Target Amount

0

Duration

Select Duration

Upload Image

Cancel

Create

Fig.No: 9.1 Creating a Campaign Page

On the crowdfunding platform, a user may launch a campaign to generate money for a project or cause. The user enters information while creating the campaign, including the project description, funding target, length of the campaign, and any potential benefits or prizes for donors.

Contributing a Campaign Page



Lung Donation

Lung donation is a profound gift, offering breath and life to those in need, a testament to the generosity that echoes long after one's last exhale.

Created By :
0x93d8f34963175...347b5F68aA

Target : 0.00001 ETH

Raised 0.0 ETH

Ends on : 4/6/2024

Contribute to this Campaign

WALLET ADDRESS	AMOUNT	DATE
0x93d8f349631757D35b81fCdd4583A5347b5F68aA	0.0 ETH	4/2/2024

Fig.No: 9.2 Contributing a Campaign Page

Funds can be contributed to the campaign by other users who are interested in helping the initiative. Each campaign typically has its dedicated page containing comprehensive information about the project's goals, funding target, duration, and associated rewards or perks for contributors. Once users select a campaign they wish to support, they proceed to contribute funds using cryptocurrency tokens native to the platform. This contribution is processed as a transaction on the blockchain network, involving details such as sender and recipient wallet addresses and the transfer amount. The transaction interacts with the smart contract governing the campaign, which automatically verifies and updates the funding status accordingly.

Withdrawal of Funds

Withdrawal Of Funds

Wallet address : 0x93d8f349631757D35b81fCdd4583A5347b5F68aA

ETH Balance :0.01

0.01

ETH

Cancel

Withdraw Funds

Fig.No: 9.3 Withdrawal of Funds Page

The campaign creator may initiate the money withdrawal process when the campaign either reaches its funding target or when its duration ends, as per the current guidelines established in the smart contract. This withdrawal procedure is automated and executed by the smart contract itself, ensuring compliance with the predefined rules and protocols governing the campaign.

CHAPTER 10

CONCLUSION AND SCOPE FOR FUTURE WORK

10.1 CONCLUSION

In conclusion, crowdfunding based on blockchain technology has become a creative and promising means of raising money for a range of enterprises and social causes. By providing a decentralised, transparent platform for direct contributions and doing away with the need for middlemen, blockchain technology transforms crowdfunding. Smart contracts lower the risk of fraud by ensuring the equitable and transparent allocation of funds. Crowdfunding campaigns may reach a wider audience and garner more donations by utilising blockchain, and the withdrawal procedure guarantees that money are utilised for the intended purpose. Crowdfunding initiatives powered by blockchain have the power to revolutionise social cause fundraising by fostering accountability and transparency. As blockchain technology becomes more widely used, we expect to see more creative crowdfunding projects that take use of its potential to have a beneficial social impact.

10.2 FUTURE ENHANCEMENT

The potential future enhancement for blockchain-driven crowdfunding platforms could involve integrating advanced identity verification mechanisms. While blockchain technology offers inherent security features and transparency, ensuring the authenticity of project creators and contributors remains crucial. By incorporating robust identity verification protocols, such as biometric authentication or decentralized identity solutions, crowdfunding platforms can enhance trust and mitigate risks associated with fraudulent activities.

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